

Smart Fake Review Detection with NLP, BERT & Behaviour Analysis

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Abstract—The rapid expansion of e-commerce has resulted in a parallel rise of fake reviews, which mislead consumer perception, damage platform integrity, and affect purchasing decisions. Conventional models for detecting fake reviews are frequently constrained by their dependence on general language patterns, insufficient contextual awareness, and ineffectiveness in integrating reviewer behavioural characteristics. The paper examines recent progress in the identification of fake reviews, focusing on the incorporation of Natural Language Processing (NLP) methodologies such as BERT and RoBERTa alongside behavioural attributes and deep learning (DL) frameworks. The models researchers proposed demonstrate significant accuracy and resilience. The paper identifies the existing gaps and proposes future research directions for explainable models and API-enabled real-time detection.

Keywords— *Fake Review Detection, Bert, RoBERTa, Behavioural Features, Deep Learning, Natural Language Processing, Explainable AI*

I. INTRODUCTION

In recent years, the growth of online marketplaces such as Amazon, Flipkart, and eBay has made user reviews essential for determining consumer responses and influencing market turnover. However, it has also created a channel for fake reviews, spreading incorrect or misleading information to mislead consumers and damage trust in digital commerce platforms [1][3][5]. Over 90% of consumers examine at least one review prior to making a purchase, and 71% consistently verify evaluations of local enterprises [23]. Research shows that a minimum of 50% of consumers put a lot of trust in reviews prior to making a purchase, establishing them as a foundation for manipulation [3][6]. 47 percent of ratings on several prominent websites have been classified as suspicious [24].

The initial studies on fake review detection directly utilized a supervised machine learning framework. Improvements to these methodologies included n-grams, TF-IDF, and LDA for modelling; however, these methods remained platform-dependent [4][9]. Advanced research [7][11][16] integrated more sophisticated notions of time and structural dynamics into models in order to enhance model-driven decision-making predictive behavioural patterns. Over 50% of shoppers claim they would abandon a purchase if they find any evidence of fake reviews [25][26].

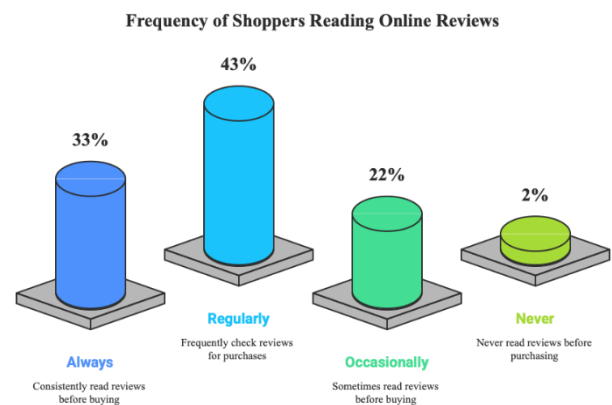


Fig. 1. Frequency of Shoppers Reading Online Review

The projection of fake reviews in 2025 anticipates global losses in online purchases to be between 770.7 and 787 billion [23]. Such a figure is stunning when considering the damage inflicted on the consumer base [Fig 1]. Alarming forecasts of the economic future suggest global spending will reach 1.07 to 1.1 trillion by 2030 [23]. The rate of spending growth coupled with the rate of fake reviews growing 12.1% faster than all internet reviews is indicative of the lagging capabilities of current strategies in place to identify and counteract hostile entities [24][27].

Among other deep learning techniques, BERT, LSTM, and CNN have successfully discerned behaviour that other techniques have failed to do. For example, BERT-like models have no trouble uncovering the clouds of the texts and reviews, irrespective of the trust that may be posted on the reviews [14][17][18]. These models usually demonstrate poor performance on the test set due to their propensity to over fit, and significant differences in style, tone, and features of the platform [5][10][21]. As well as this, online detection of deceptive reviews remains an unresolved problem stemming from substantial volumes of data and the need for rapid response and data analysis [20][22][28].

The public domain lack accessible datasets with adequate annotations is a monumental challenge. Public datasets used for research are often outdated, noisy, or limited in scale, which undermines genuine evaluation alongside practical implementation [6][8][19]. Hence, more collaborative efforts are required from the research community in order to

develop benchmark datasets that cover a wide range of reviews across different platforms and languages.

A. Research Questions

The authors seek to address three key research questions:

R.Q.1. How can user behavioural features (e.g., review frequency, timing, polarity distribution) enhance fake review detection?

R.Q.2. What methods can improve model explainability in classifying fake vs. genuine reviews?

R.Q.3. How can real-time detection be achieved through API-based or live monitoring systems?

B. Structure of the paper

The body of work has been compartmentalized into five distinct head sections or parts. In Section I, the reader is able to find an exhaustive framework on review fraud on e-commerce platforms. This framework highlights the pivotal role of the review fraud, its importance, and the objectives of the research. Section II outlines the literature review, which synthesizes research, which has been conducted on the review fraud identification, models constructed, and current gaps which persist, particularly in regard to the gaps in absence of explainability, real-time functioning, and the behavioural features. In Section III the authors made an attempt to devise and outline a methodological approach which included the selection of relevant datasets, defining features which need to be captured, the determination of classifier models, and archiving system architectural diagrams. Section 4 encompasses the outcomes and the juxtaposition of the outcomes with past models and the review fraud on the e-commerce platforms. These conclusions synthesize the major findings of the study and their relevance in the context of advancing more open and credible review systems on the internet.

II. LITERATURE REVIEW

M. Borghi et al. [1] studied the role of Machiavellianism, narcissism, and psychopathy concerning the review fraud phenomenon on the internet. They utilize the Interpersonal Deception Theory to help them sift through a large corpus of Amazon product reviews, looking for linguistic markers indicative of the target personality traits. There is a disproportional correlation between fake reviews and Machiavellianism. One Machiavellian comments, Machiavellians employ fake reviews as a means of social engineering to sway people's opinions on products, while narcissists, on the other hand, do so in order to use their reviews to enhance their perceived social status, as research suggests. The authors encourage the exploration of personality traits systems of false review detection, as well as, their stability and influence on the online domain.

Using the RF classifier and opinion mining, K. Mane et al.'s [2] study focuses on the identification of the fake purchase reviews. The authors performed tokenization, stop words omission, and feature extraction on a dataset of 90,000 reviews corresponding to Amazon to refine detection. The study, as mentioned, attained an accuracy of 79.47%, which improved with the increase in the size of the

dataset. Text-based attribute classification improves with the incorporation of sentiment analysis. The collection of user behaviour graphed with the changing patterns of copy review facerequires the construction of more diverse and larger datasets.

In the study by K. Mane et al. [3], ML algorithms are used for the detection of fake reviews on e-commerce sites. The authors examine the use of basic classifiers, deep learning techniques, and various transformer models, such as BERT. The need for improved accuracy in detection motivated the integration of opinion mining and sentiment analysis with the language and user behaviour factors. It is the authors' contention that more effective identification of false reviews can be achieved by employing ensemble and hybrid strategies which integrate multiple machine and deep learning models.

C. Singh et al. [4] applies ML to detecting fake reviews on online shopping sites via sentiment analysis and n-grams derived from reviews. The authors present a trained model augmented with classifiers and advanced feature engineering techniques on a large dataset of product reviews. The Random Forest model, on the other hand, outperforms SVC and LoR due to its ensemble learning technique of integrating multiple decision trees. The study also integrates language and sentiment with metadata to demonstrate the potential of detecting fake reviews.

J. Liu et al. [5] propose a new approach for detecting fake reviews using the massive language model RoBERTa's contextual text representation and user behaviour. The authors address standard detection tactics that are challenged by text variances, language differences, and reviewer disguise. Using RoBERTa's self-attention-based embeddings with behavioural signals like activity window, review order, and rating manipulation, the hybrid model collects linguistic and behavioural signals. To overcome the imbalance between true and false reviews, SMOTE data augmentation improves model robustness. The RoBERTa-LSTM-CNN model with merged behavioural features outperforms standard ML classifiers.

R. Mohawesh et al. [6] propose a semantic- and linguistic-sensitive model for detecting fake reviews using the RoBERTa transformer and an LSTM layer. The authors introduce a hybrid technique that uses RoBERTa's deeply contextualised embeddings and LSTM's long-range dependencies to address deficiencies in previous language feature-focused approaches. Experimental results on benchmark datasets reveal that the recommended model outperformed existing strategies by a wide margin with 96.03% and 93.15% accuracy, respectively. The research uses explainability methods like SHAP to provide model choice insights and boost transparency and confidence.

A hybrid DL model for detecting fake reviews by R. A. Duma et al. [7] effectively uses review text, overall rating, and aspect rating for improved classification. The authors propose integrating contextualised word embeddings from BERT with aspect rating and overall rating. The model handles semantic and sequential review text characteristics with CNN and LSTM layers, while aspect ratings provide fine-grained satisfaction information to complement overall

evaluations. The proposed technique outperforms, reaching 99.5% accuracy. The model has a better forecast time, making it appropriate for real-time applications.

P. Gupta et al. [8] use transfer learning and transformer-based pre-trained models. The research overcomes the drawbacks of typical ML and neural network models to build a generalised and effective classifier for small labelled datasets and low processing resources. The authors preprocess the Yelp dataset using real and false reviews without losing semantic significance by avoiding stop word removal. Models were fine-tuned using 10% and 50% dataset segments and tested. RoBERTa 50% data outperformed BERT, ALBERT, and DistilBERT. The study also compares computational training time and finds that DistilBERT balances accuracy and efficiency.

Y. Fang et al. [9] propose a dynamic knowledge graph-based approach (DKG-FRD) to identify fake reviews by analysing detailed relationships between reviewers, reviews, products, and stores. A sophisticated sentence vector twin-word embedding conditioned bidirectional LSTM (ST-BLSTM) architecture is used to extract information accurately, and a dynamic knowledge graph with time-series features captures review evolution over time. Experimental results on Amazon, Netflix, and MovieLens show that the DKG-FRD technique outperforms baseline approaches with accuracies over 93%.

R. Mohawesh et al. [10] propose an ensemble model of three transformer-based models, RoBERTa, XLNet, and ALBERT, to extract strong contextualised representations from review texts to improve fake review detection. Removing noises, tokenising, and fine-tuning transformer models with classifier heads for binary classification are significant preprocessing steps. The ensemble model outperforms transformers in accuracy on two benchmarking datasets, OpSpam and Deception. It also performs well on tiny datasets.

M. Puttarattanamanee et al. [11] compare sentiment analysis methods for spotting fake online purchasing reviews. The authors evaluate GPT-2, NBSVM, BiLSTM, and RoBERTa using Amazon Review Data 2018. Special character removal, tokenisation, Bag of Words, and TF-IDF word representation were used for strict preprocessing. To maximise model accuracy, hyperparameters were optimised. RoBERTa outperformed all other models with 97% accuracy, followed by NBSVM at 95%, BiLSTM at 92%, and GPT-2 at 82%. RoBERTa has the best AUC score of 0.9976, indicating higher accuracy in identifying fake reviews.

S. M. Abd- Alhalem et al. [12] utilizes DL and aspect-based sentiment analysis for fake e-commerce reviews detection. Reviews are analysed to extract deep aspect of documentation features extracted using CNN, tagged with PoS, and GloVe embedded analytic vectors. Classifiers are trained over a seamless network. The model is subjected to a rigorous evaluation of Amazon reviews and earns a high score of 97.73% accuracy. The paper captures the increased correctness of fake reviews detection due to fusion of lexical and semantic features and reiterates the need to analyse the complexities in fake reviews.

N. M. Gardazi et al. [13] reports on BERT for NLP and its new capability. BERT and its descendants enjoy a continual edge over traditional models owing to the capacity for contextual embedding. The authors also addresses the expenditure of computing resources, the easiness of finetuning, the need of domain and interpretable adaptation, and the reduction of bias.

J. Lu et al. [14] constructed their BSTC fake review detection model with CNNs and pre-trained language models for easy fake review classification and detection. This model integrates BERT for feature semantics, SKEP for sentiment knowledge, and TextCNN for local feature extraction. After rigorous testing spanning three standard benchmark datasets, the BSTC model achieved accuracy levels between 91.25% and 93.44% and outperformed several other DL models.

S. N. Alsubari et al. [15] take a different approach and use supervised ML techniques to identify fake reviews. They work with the fake hotel reviews dataset from TripAdvisor and apply complex preparation techniques like punctuation and stop word removal and tokenization. The authors further enhance the text representation through TF-IDF feature extraction and part-of-speech tagging and sentiment ratings with n-grams. They test SVM, NB, RF, and AdaBoost classifiers for accuracy to enhance results. The authors independently evaluate their approach with related literature and note the superiority of their findings.

Detecting fake reviews using the psycholinguistic framework and transformer-based NLP models was accomplished by J. Salminen et al. [16] Psycholinguistic analysis and transformer layers of psycholinguistic and demographic classifiers were computed and analysed over seven demographic, socio-economic, and psychographic groupings of 3,070 UK-representative, 5 commodity classes clustered with half-true and half-false reviews. Authentic versus simulated reviews showed, through LIWC, enhanced and increased psycholinguistic differences, including emotional and analytical tones, versus negative emotional and cognitive simulator tones. BERT, RoBERTa, and XLNet transformer models surpass traditional classifiers in all product and sentiment quadrants.

Roja D. et al. [17] developed an AI system equipped with deep learning algorithm and sentiment classifier capable of detecting and analysing hotel reviews as well as false reviews. BERT and LSTM-based models were employed in the system to sort consumer feedback as positive, negative, or neutral. CNN in combination with behaviour analysis was utilized in the research to pinpoint fraudulent reviews. With XGBoost and LSTM models, the platform predicts hotel business performance with an error tolerance of less than 5%. The research showed that artificial intelligence-based review analysis for hotels assists in optimizing the rates, service, and marketing, as well as customer satisfaction, resulting in business growth.

P. Phukon et al. [18] introduced a technique called FakeDetectionGCN which sought to identify fake reviews using aspect based sentiment analysis (ABSA) and graph convolutional networks. The technique treats each review as a graph where certain words, or aspect keywords, function

as nodes and certain semantic dependencies function as edges and are assigned weight using SenticNet emotion. The technique uncovers real from false reviews by capturing complex attitude-aspect interactions. Using OTT and Salminen as benchmark datasets, the technique shows consistent performance by achieving 92.33% accuracy and 0.97 AUC.

A. M. Elmogy et al. [19] constructed a supervised ML technique which aims to detect fake reviews through textual data and reviewer behavioural data. The data set consists of 5,853 Yelp hotel reviews of which 1,144 are fake and the rest, 4,709, are real. The authors employ certain NLP approaches to preprocess the data. The feature extraction process which quantifies reviewer behaviour uses TF-IDF bi-gram and tri-gram models and other artificially generated behavioural features such as the use of capital letters, punctuation, and emojis. The study aims to determine the optimal classifier from the five classifiers available and concludes that K-Nearest Neighbour (KNN) with K set to 7 has the highest F1 score.

Abrar Qadir Mir et al. [21] use supervised machine learning classifiers and the BERT model to obtain complex word embeddings from review text to detect online fake reviews. The authors compare their strategy using popular hotel, restaurant, doctor, and Amazon databases. BERT preprocesses text data into word embeddings and trains several classifiers. In hotel dataset experiments, SVM has the highest accuracy of 87.81%. The study shows that text content alone can distinguish false reviews from real ones without sentiment analysis.

M. Tabany and M. Gueffal [22] use an SVM supervised learning model to analyse Amazon reviews for sentiment and fake reviews. Hyperparameter adjustment improves SVM sentiment analysis accuracy to 93%. Shorter reviews improve classifier accuracy. An artificially generated false reviews dataset created using GPT-2 and evaluated for authenticity shows that fake reviews average 46.3 words and authentic reviews 54.75 words. Unlabelled data affects supervised learning models; hence, semi-supervised or unsupervised methods are recommended.

III. PROPOSED METHODOLOGY

In e-commerce reviews systems, detecting fakes starts with gathering a dataset, which is followed by preprocessing and feature selection centred on relevant information for model development. Training a detection model is next, using an API for model modularization and scalability for real-time review processing. Once trained, the model on the other hand, and the methodology moves to the emerging phase, which model focuses on assessing the obtained results and deriving the appropriate conclusions. The last phase is the collection and analysis of the actionable results, which in turn, provide the frameworks for the platform operators and other participants. Figure 2 contains the data flow diagram of the proposed methodology.

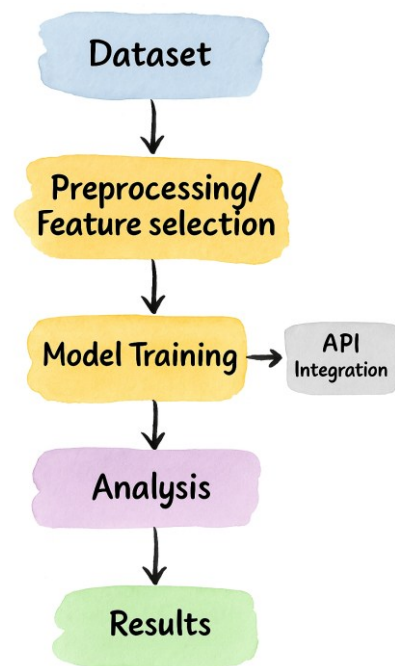


Fig. 2. Data Flow Diagram of the proposed methodology

A. About Dataset

The research was conducted based on the dataset of the reviewed products on the Amazon platform. The dataset consists of reviews for diverse products, such as Electronics, Clothing, and Books. Every time a rating is retrieved, paired with the review is a collection of text and behaviour information. The review consists of its entirety, abstract form, and the star rating system. In addition to this, user metadata is included, which includes the ID of the reviewer, whether the review is verified, the review's helpfulness score, the review's publish date, and the summary of user reviews. Overall, this collection of variables offers a solid data structure for the combination of the evaluation context and content.

B. Data Preprocessing and Feature Selection

In preprocessing step, review text is cleaned by lowercasing all characters, removing stop words, and tokenizing sentences, as well as stemming and lemmatizing words. Also, behavioural aspects are extracted such as time gaps between subsequent reviews (in order to capture those reviews which are posted immediately after the availability of the product and which are suggestive of rapid submission), explosive and stagnant periods of review, the verified-purchase status of the reviews as a trust metric, and sentiment shifts across multiple reviews from a single reviewer. Textual features are vectorized and normalized with TF-IDF and behavioural features are pruned with correlation thresholds to retain only the most predictive features.

C. Model Training

After the feature engineering step, various models are fitted and compared. Conventional ML-based classifiers are used as strong baseline models. Other than that, the LSTM model intelligently extracts sequential knowledge within reviews, and a pre-trained BERT transformer model captures deep semantic relations among the review texts.

All models are evaluated by accuracy, precision, recall, and F1-score. For qualitative measures, the authors evaluate the robustness of the model through cross-category tests and determine the extent to which the model can generalize across varying product categories.

D. API Inegration

To have access to real-time fake review detection, the authors combine the trained models with the Grok API. This API employs large language models to analyse and categorize new reviews as fake and genuine. It returns not only a class label but also a human-interpretable explanation, which leads to greater interpretability. This API layer integration successfully turns the offline detection system into a live service with sub-second latencies and is demonstrated to be effective on a live e-commerce site.

IV. RESULT & ANALYSIS

Grok's explainability arsenal, in tandem with SHAP (SHapley Additive exPlanations), computes the factors which drove the model's decisions. The framework, then, is capable of understanding which types of behavioural and text features are strongly predictive by comparing outcomes across product types. Also, and more interestingly, the authors do detailed error analysis on the reviews which the model classifies incorrectly to better the model in the future.

The procedures prescribed solely for obtaining detections of fake reviews are proficient as they note the red flags of promotional phrases intersectional to irregular posting patterns as well as the absence of verifiable transactions and remove these reviews systematically. All other reviews are genuine and show more user emotion, actual engagement, and verifiable transactions. The system performs exceptionally well with both falsification and affirmation negations for the reviews to maintain the proposed user strategy, which strongly increases the system's utility and trustworthiness for platform administrators as well as end users. The derived accuracies are shown in Table 1 along with comparison to the other available models. A comparison of the models is shown in figure 3.

TABLE I. MODEL COMPARISON

Year of Publication	Authors	Methods Used	Accuracy
2025	K. Mane et al. [2]	RF, Opinion Mining	79.47%
2024	K. Mane et al. [3]	SVM, CNN+PSO, LSTM, BERT	99.4%
2024	R. Mohawesh et al. [6]	RoBERTa + LSTM	96.03%

Year of Publication	Authors	Methods Used	Accuracy
2021	P. Gupta et al. [8]	BERT, RoBERTa, ALBERT, DistilBERT	69%
2020	Y. Fang et al. [9]	ST-BILSTM, DKG	93.41%
2021	R. Mohawesh et al. [10]	RoBERTa + XLNet + ALBERT (ensemble)	94.37%
2023	M. Puttarattanam anee et al. [11]	RoBERTa, BLSTM, GPT-2, NB/SVM	97%
2025	N. M. Gardazi et al. [13]	BERT	90%
2023	J. Lu et al. [14]	BERT, SKEP, TextCNN	93.44%
2022	S. N. Alsubari et al. [15]	RF, SVM, AdaBoost, NB, POS, sentiment features	95%
2025	J. Salminen et al. [16]	LIWC, BERT, RoBERTa, XLNet	82.6%
2025	D. Roja et al. [17]	BERT, LSTM, CNN + behavioural features	91%
2025	P. Phukon et al. [18]	BLSTM, GCN	92.33%
2021	A. M. Elmogy et al. [19]	KNN, SVM, Naive Bayes, LR, RF, TF, behavioural features	87.87%
2023	A. Q. Mir et al. [21]	BERT, SVM, RF, AdaBoost	87.81%
2024	M. Tabany et al. [22]	SVM, Naive Bayes, LR, RF, NLP preprocessing	88%

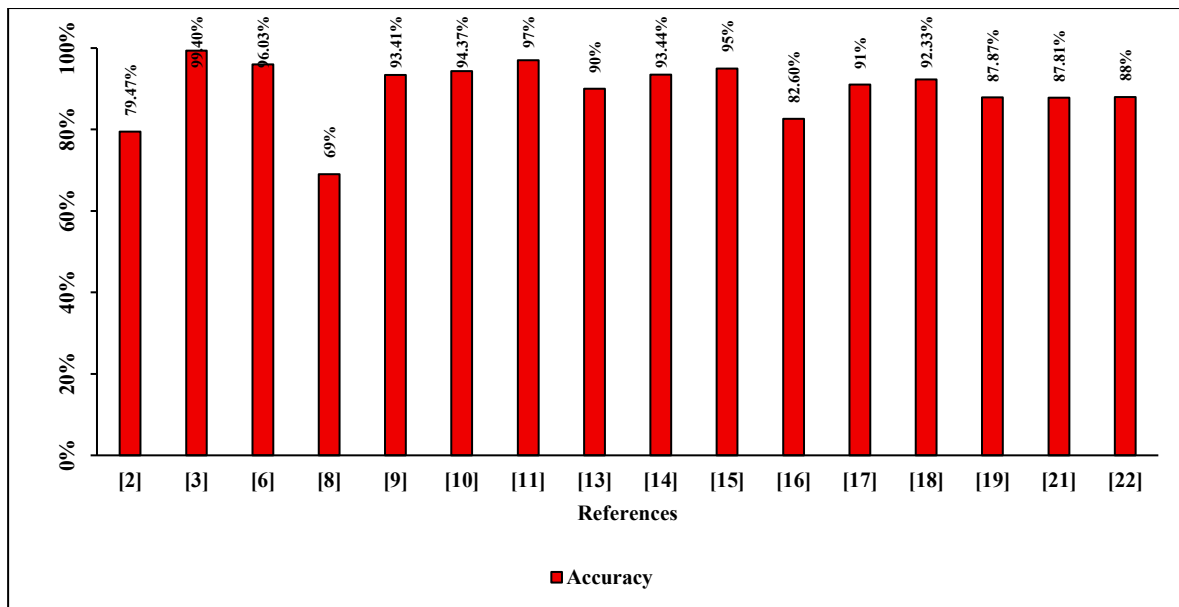


Fig. 3. Comparison Graph

V. CONCLUSION

This study presents an overview of the current advancements in hybrid strategies for detecting fake reviews, integrating NLP with ML models while also considering user behaviour. The integration of review content with behavioural indications, such as clicking behaviour and the "verified purchase" indicator, significantly improves detection accuracy. Transformer models like BERT outperform conventional classifiers when applied to authentic Amazon data. The authors utilize the Grok API for internet filtration, which detects fake reviews and explains each determination. This combined approach achieves superior accuracy, is suitable for real-time review screening, and offers clear, interpretable justifications for classification, hence enhancing its practicality and reliability for live e-commerce platforms.

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